

LAYER 1: DATA PRODUCT FACTORY

CRM (1.02M Custs)
crm_customer_id, PII Profile
Cards (660k Accounts)
card_account_id, src_cust_ref
Lending (404k Loans)
loan_id, borrower_number
Core Banking (780k Dep)
deposit_account_id, 10-digit CIF
Collections (367k Cases)
case_id, coll_customer_ref

ENTITY RESOLUTION ENGINE
- Deterministic Matching (Phone/Email/DOB = 1.0)
- Probabilistic Matching (Jaro-Winkler >= 0.92)
- Review Queue for Confidence < 0.85 (10,365 records)

GOLDEN FINANCIAL ID (GID-XXXXXXXX)
1,001,879 Unique Borrower Identities Resolved

Vectorized Analytical Lakehouse
DuckDB In-Process Engine (68,162,322 records)
18 Automated DQ Rules (0 critical defects)
ODCS v3.0.0 Data Contract (DC-COLL-360)

GOLDEN CUSTOMER 360
Single Governed Table (golden_customer_360)
Zero Protected Attributes Full Lineage Traceability
Parquet Lakehouse Export (1,001,879 rows in 8.4s)

LAYER 2: INSIGHT & NLP PLATFORM

Natural Language to SQL Engine
- Translates supervisor / agent queries to DuckDB SQL
- Always displays executed SQL alongside query results
- Evaluated on 35 Official Benchmark Scenarios
Refusal & Privacy Guardrails
- Refuses Out-of-Scope queries (BQ-018: Weather)
- Refuses Bulk PII Exfiltration (BQ-023: Ontario list)
- Refuses Protected Attributes (BQ-029/031: Gender/FSA)
35 Benchmark Question Evaluation
100% Benchmark Coverage (Dev + Test Splits)
BQ-035 Open Cases by Bucket: Exact Integer Match

LAYER 3: COLLECTION MEMORY & FEATURES

Longitudinal Collection Memory
- 3,581,511 Historical Interactions & PTPs
- NLP Hardship Detection: 84,289 notes flagged
- Solves the Five Golden Questions instantly
3-Branch Financial Health Engine
Branch A ■ Exposure Analysis (DTI Ratio)
Branch B ■ Cash-Flow Stress (Liquidity & Lag)
Branch C ■ Payment Behaviour (PTP Rate & DPD)
Debt Service Ratio (DSR) & Affordability Assessment
1,001,879 Customers Scored in 0.96s (DuckDB SQL)
Composite: GOOD (895k) FAIR (45k) STRESSED (43k) CRITICAL (19k)
Can Support Arrangement: YES / MARGINAL / NO

LAYER 4: AI DECISIONING & AGENT ASSIST COCKPIT

POLICY-CONSTRAINED NEXT BEST ACTION (NBA)
93,240 Active Delinquency Recommendations Evaluated in 0.88s
Rule 1: Hardship / Vulnerability -> Specialist Review (Protection)
Rule 2: DPD > 90 -> Senior Escalation & Review
Rule 3: Sustainability Gate (Cannot pay) -> Hardship Assistance
Rule 4: Marginal Financial Capacity -> Structured Payment Plan
Rule 5: Early Stage (<=30 DPD) + Healthy -> Automated Digital Reminder
Distribution: HUMAN_REVIEW: 58.5k REMINDER: 20.9k PAYMENT_PLAN: 13.8k

Agent Assist Cockpit (Streamlit Interface)
Panel 1: Golden Financial ID & Customer 360 Resolved borrower name, primary contact, cross-product exposure, delinquency history.
Panel 2: Financial Health Traffic Lights Real-time visual badges: Exposure (Low/Med/High), Cashflow, Payment Behaviour, Composite Health.
Panel 3: Collection Memory & Five Golden Questions Timestamped interaction timeline, PTP records, previous promises kept/broken, agent notes.
Panel 4: Policy-Constrained Recommendation Recommended action, plain-English explanation, factual evidence tags, cited policy rule.
Panel 5: Human-in-the-Loop Controls Frontline agent control: Accept, Modify, or Override recommendation with mandatory audit reasoning.
Panel 6: Smart Payment Plan Formulator 3 affordability-tested plan options (Fast-Track, Balanced, Step-Up) calculated from monthly surplus.
Panel 7: Grounded Natural Language Query Ad-hoc portfolio exploration with verified DuckDB SQL generation and table visualization.

CROSS-CUTTING GOVERNANCE, AUDITABILITY & FAIRNESS
Zero Protected Attributes in Decision Models Full Audit Trail Logging Human Override Rights
Deterministic Math (LLM is Never Calculator of Record) CASL & Canadian Regulatory Compliance
Production Performance: End-to-End Execution of 68M Records in Under 70 Seconds